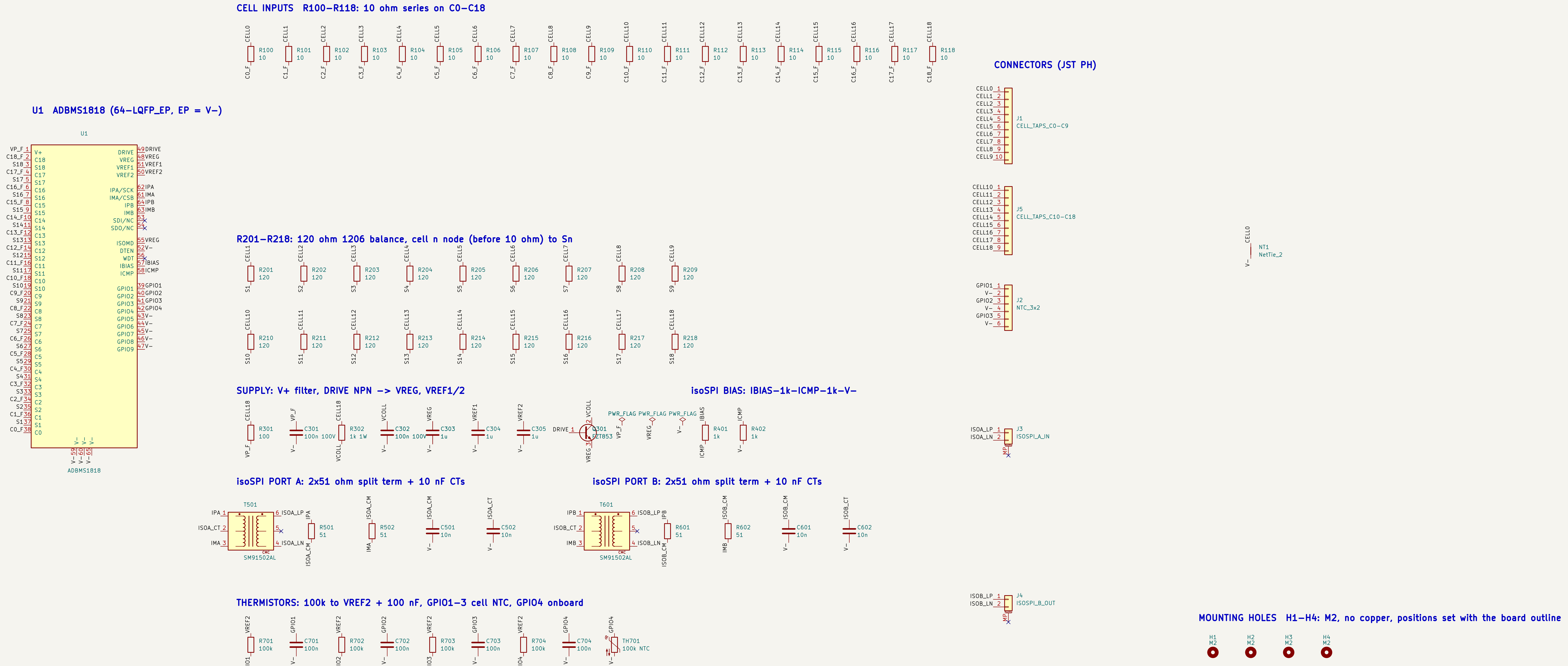
Wired with net labels: labels with the same name are one net. 18S Lean: no fuses, clamps or C-pin caps; generated from slave-18s by make_lean.py, do not edit by hand. C0 sense (CELL0, J1.1) and V- meet only at net tie NT1, which replaces F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave-18s-design-basis.md.



**CELL INPUTS  R100–R118: 10 ohm series on C0–C18**

**U1  ADBMS1818 (64-LQFP_EP, EP = V-)**

**R201–R218: 120 ohm 1206 balance, cell n node (before 10 ohm) to Sn**

**SUPPLY: V+ filter, DRIVE NPN -> VREG, VREF1/2**

**isoSPI BIAS: IBIAS–1k–ICMP–1k–V-**

**isoSPI PORT A: 2x51 ohm split term + 10 nF CTs**

**isoSPI PORT B: 2x51 ohm split term + 10 nF CTs**

**THERMISTORS: 100k to VREF2 + 100 nF, GPIO1–3 cell NTC, GPIO4 onboard**

**CONNECTORS (JST PH)**

**MOUNTING HOLES  H1–H4: M2, no copper, positions set with the board outline**

CELL0 R100 10 C0_F
CELL1 R101 10 C1_F
CELL2 R102 10 C2_F
CELL3 R103 10 C3_F
CELL4 R104 10 C4_F
CELL5 R105 10 C5_F
CELL6 R106 10 C6_F
CELL7 R107 10 C7_F
CELL8 R108 10 C8_F
CELL9 R109 10 C9_F
CELL10 R110 10 C10_F
CELL11 R111 10 C11_F
CELL12 R112 10 C12_F
CELL13 R113 10 C13_F
CELL14 R114 10 C14_F
CELL15 R115 10 C15_F
CELL16 R116 10 C16_F
CELL17 R117 10 C17_F
CELL18 R118 10 C18_F
U1
VP_F 1 V+
C18_F 2 C18
S18 3 S18
C17_F 4 C17
S17 5 S17
C16_F 6 C16
S16 7 S16
C15_F 8 C15
S15 9 S15
C14_F 10 C14
S14 11 S14
C13_F 12 C13
S13 13 S13
C12_F 14 C12
S12 15 S12
C11_F 16 C11
S11 17 S11
C10_F 18 C10
S10 19 S10
C9_F 20 C9
S9 21 S9
C8_F 22 C8
S8 23 S8
C7_F 24 C7
S7 25 S7
C6_F 26 C6
S6 27 S6
C5_F 28 C5
S5 29 S5
C4_F 30 C4
S4 31 S4
C3_F 32 C3
S3 33 S3
C2_F 34 C2
S2 35 S2
C1_F 36 C1
S1 37 S1
C0_F 38 C0
DRIVE 49 DRIVE
VREG 48 VREG
VREF1 51 VREF1
VREF2 50 VREF2
IPA/SCK 62 IPA
IMA/CSB 61 IMA
IPB 64 IPB
IMB 63 IMB
SDI/NC 53
SDO/NC 54
ISOMD 55 VREG
DTEN 52 V-
WDT 56
IBIAS 57 IBIAS
ICMP 58 ICMP
GPIO1 39 GPIO1
GPIO2 40 GPIO2
GPIO3 41 GPIO3
GPIO4 42 GPIO4
GPIO5 43 V-
GPIO6 44 V-
GPIO7 45 V-
GPIO8 46 V-
GPIO9 47 V-
V- 59 V-
V- 60 V-
V- 65 V-
ADBMS1818
CELL1 R201 120 S1
CELL2 R202 120 S2
CELL3 R203 120 S3
CELL4 R204 120 S4
CELL5 R205 120 S5
CELL6 R206 120 S6
CELL7 R207 120 S7
CELL8 R208 120 S8
CELL9 R209 120 S9
CELL10 R210 120 S10
CELL11 R211 120 S11
CELL12 R212 120 S12
CELL13 R213 120 S13
CELL14 R214 120 S14
CELL15 R215 120 S15
CELL16 R216 120 S16
CELL17 R217 120 S17
CELL18 R218 120 S18
CELL18 R301 100 VP_F
VP_F C301 100n 100V V-
CELL18 R302 1k 1W VCOLL
VCOLL C302 100n 100V V-
VREG C303 1u V-
VREF1 C304 1u V-
VREF2 C305 1u V-
DRIVE 1 Q301 FZT853 2 VCOLL 3 VREG
PWR_FLAG VP_F
PWR_FLAG VREG
PWR_FLAG V-
IBIAS R401 1k ICMP
ICMP R402 1k V-
T501
IPA 1 6 ISOA_LP
ISOA_CT 2 5
IMA 3 4 ISOA_LN
CMC
SM91502AL
IPA R501 51 ISOA_CM
ISOA_CM R502 51 IMA
ISOA_CM C501 10n V-
ISOA_CT C502 10n V-
T601
IPB 1 6 ISOB_LP
ISOB_CT 2 5
IMB 3 4 ISOB_LN
CMC
SM91502AL
IPB R601 51 ISOB_CM
ISOB_CM R602 51 IMB
ISOB_CM C601 10n V-
ISOB_CT C602 10n V-
VREF2 R701 100k GPIO1
GPIO1 C701 100n V-
VREF2 R702 100k GPIO2
GPIO2 C702 100n V-
VREF2 R703 100k GPIO3
GPIO3 C703 100n V-
VREF2 R704 100k GPIO4
GPIO4 C704 100n V-
GPIO4 TH701 100k NTC V-
CELL0 1
CELL1 2
CELL2 3
CELL3 4
CELL4 5
CELL5 6
CELL6 7
CELL7 8
CELL8 9
CELL9 10
J1
CELL_TAPS_C0-C9
CELL10 1
CELL11 2
CELL12 3
CELL13 4
CELL14 5
CELL15 6
CELL16 7
CELL17 8
CELL18 9
J5
CELL_TAPS_C10-C18
GPIO1 1
V- 2
GPIO2 3
V- 4
GPIO3 5
V- 6
J2
NTC_3x2
CELL0 NT1 NetTie_2 V-
ISOA_LP 1
ISOA_LN 2
MP
J3
ISOSPI_A_IN
ISOB_LP 1
ISOB_LN 2
MP
J4
ISOSPI_B_OUT
H1 M2
H2 M2
H3 M2
H4 M2

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818-slave-18s-lean.kicad_sch

***Title: ADBMS1818 18S slave module – Lean (no hot-plug protection)***

Size: User
Date: 2026-11-06
**Rev: 0-draft**
KiCad E.D.A. 10.0.6
Id: 1/1